

# MATHEMATICAL PHYSICS

2016–2017

MP204

## Electricity and Magnetism

Time allowed:  $1\frac{1}{2}$  hours

Answer **ALL** questions

This is a **SAMPLE** exam, roughly reflecting the general structure of the finals for 2016 – 2017.

Remember that **ALL** questions are to be answered.

If any figures are relevant, please include them with your solution! Drawing appropriate pictures often helps you arrive at the correct solution.

1. (a) Ampere's theorem for the magnetic field due to a steady current is

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

What does  $C$  denote? What does the symbol  $I$  refer to?

Starting from this integral relation, derive part of Maxwell's fourth equation (which concerns the curl of the magnetic field). If you use a theorem about vector integrals on the way, name the theorem explicitly.

[15 marks]

- (b) A solid metallic sphere has radius  $R$ , and carries a total charge  $Q$ . The metal is an excellent conductor.

In which part of the sphere is the charge located?

Write expressions for the electric field magnitude at a distance  $r$  from the center of the sphere, both inside the sphere ( $r < R$ ) and outside the sphere ( $r > R$ ). Write expressions for the electric potential as a function of  $r$ , both inside and outside the sphere.

Sketch a plot of the electric field magnitude as a function of the distance  $r$  from the center of the sphere. Your plot should run from  $r = 0$  to  $r = 2R$ .

[15 marks]

- (c) An electromagnetic field is described by the scalar potential  $V$  and vector potential  $\mathbf{A}$ , given by

$$V = -LB_0\omega x \cos(\omega t); \quad \mathbf{A} = B_0y \sin(\omega t)\hat{i}$$

where  $B_0$ ,  $L$ , and  $\omega$  are positive constants. Find the electric and magnetic fields in this system. In which spatial positions (which part of space) does the electric field vanish?

Use Maxwell's equations to calculate the current density and the charge density in this system.

[20 marks]

2. (a) A flat square plate lies parallel to the  $x$ - $y$  plane and is charged with a variable surface charge density,  $\sigma = -4\gamma xy$ . The four corners of the square plate are at the points  $(0, 0, z_0)$ ,  $(L, 0, z_0)$ ,  $(0, L, z_0)$ ,  $(L, L, z_0)$ . Find the total electric flux through a surface that completely surrounds this charged plate. (Here  $\gamma$  is a constant.)

[17 marks]

- (b) The magnetic field in a region changes with time,  $\mathbf{B} = B_0 e^{-2t/t_0} \hat{i}$ . Here  $B_0$  and  $t_0$  are positive constants. We consider two square loops. One square loop lies in the  $y$ - $z$  plane, and has sides of length  $L_1$ . Find the electromotive force (EMF) induced in this loop. The other square loop lies in the  $x$ - $y$  plane, and has sides of length  $L_2$ . Find the EMF induced in this loop.

[15 marks]

- (c) A long solenoid has  $n$  turns per unit length and carries time-varying current  $I(t) = I_0 \sin(\omega t)$ . Use Faraday's law in integral form to calculate the induced electric field as a function of the distance  $r$  from the axis of the solenoid, both *inside* and *outside* the solenoid. Reminder: A current  $I$  through a solenoid creates a magnetic field  $\mu_0 n I$  inside the solenoid, where  $n$  is the number of turns per unit length.

[18 marks]